

S7CAS-PLEIADES-CARTE-MERE

Pleiades

01

Revision 1.00

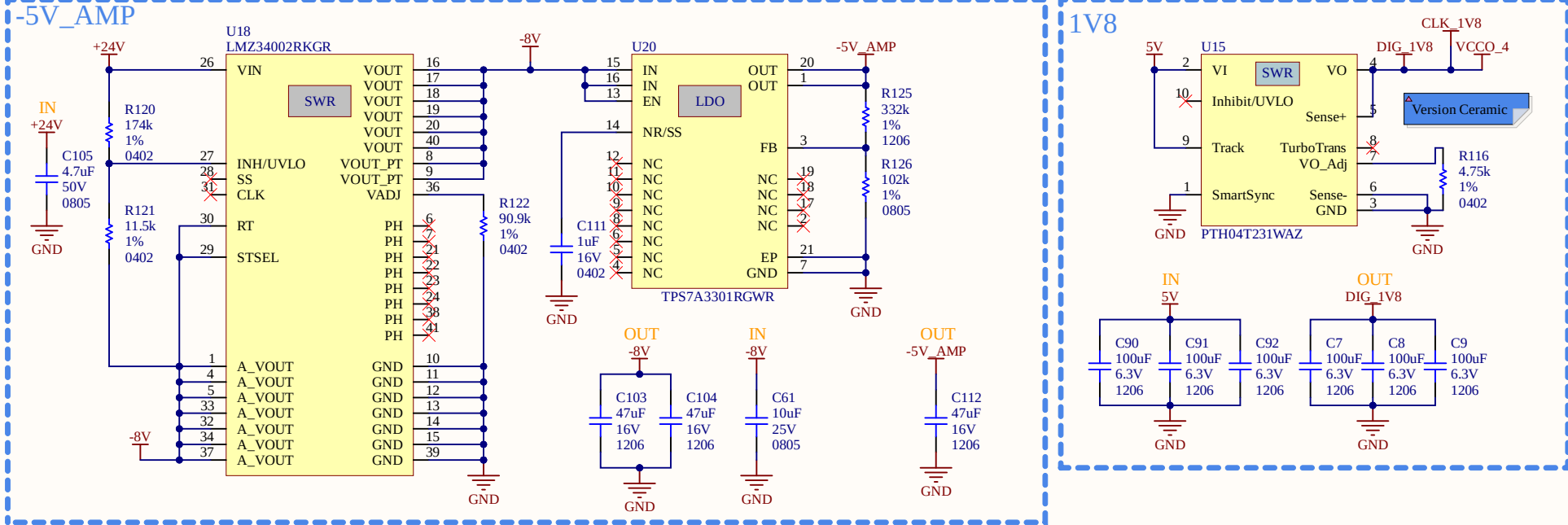
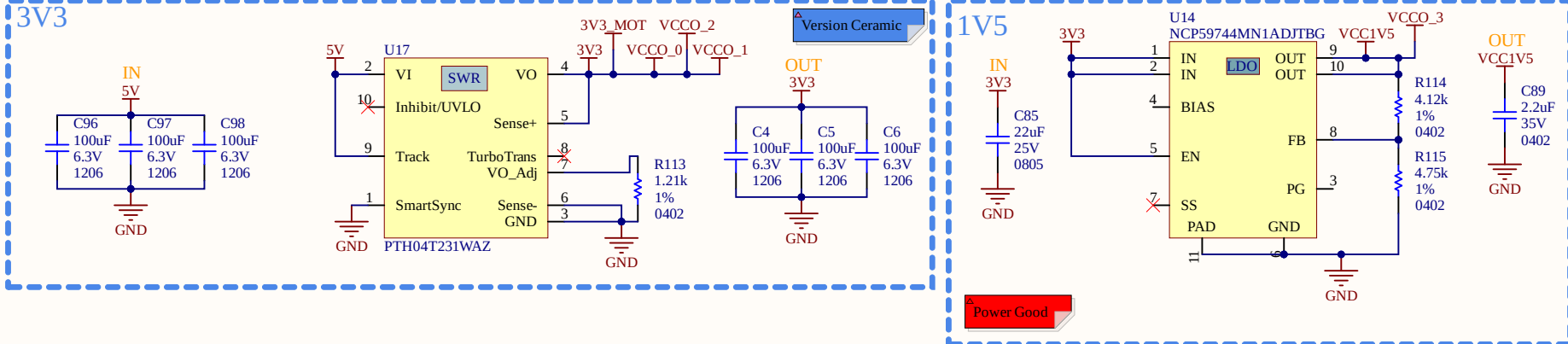
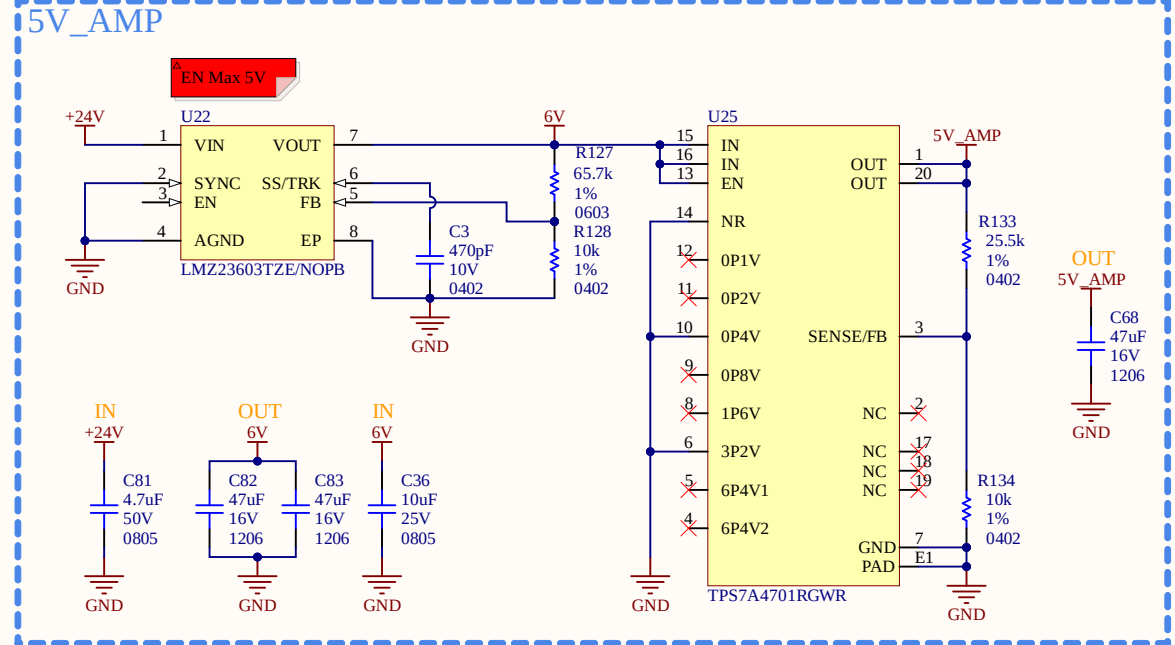
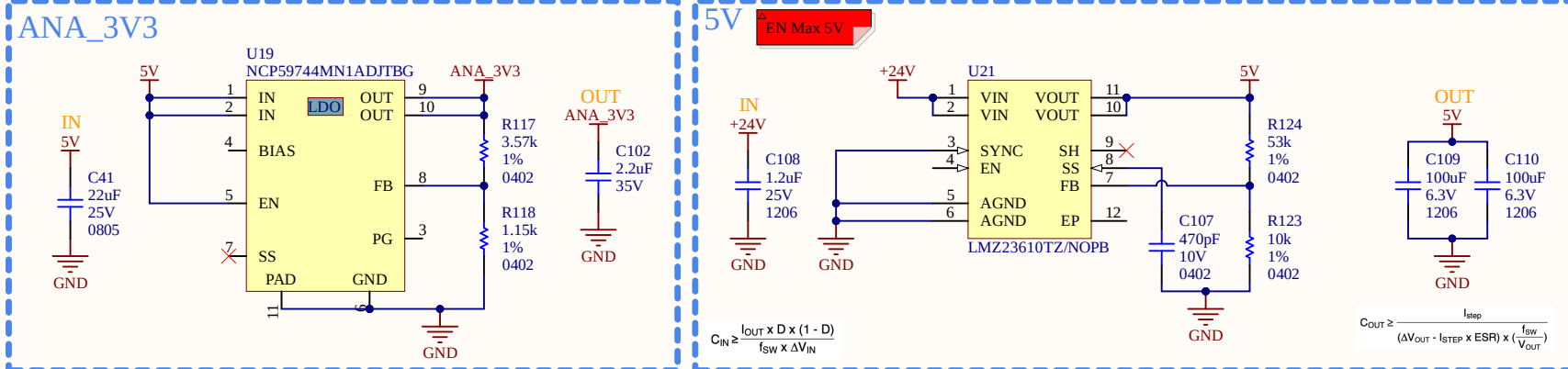
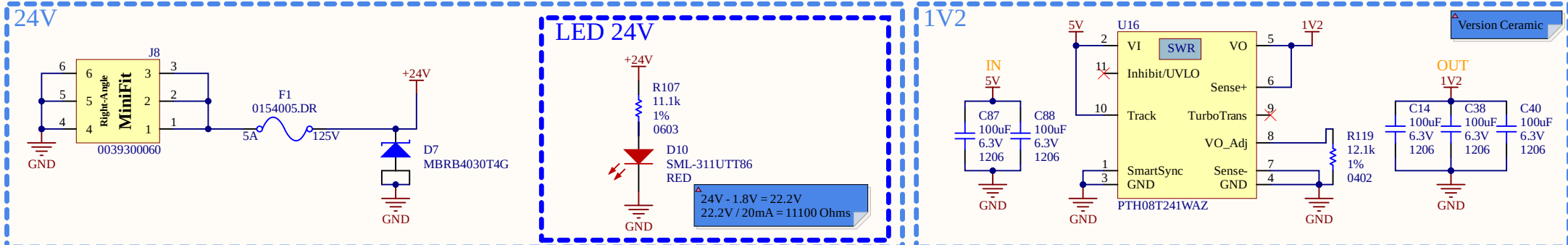
Date : 13/01/2026

Revision history	

Package size conversion	
Metric	Imperial
1005	0402
1608	0603
2012	0805
3216	1206
3225	1210
6432	2512

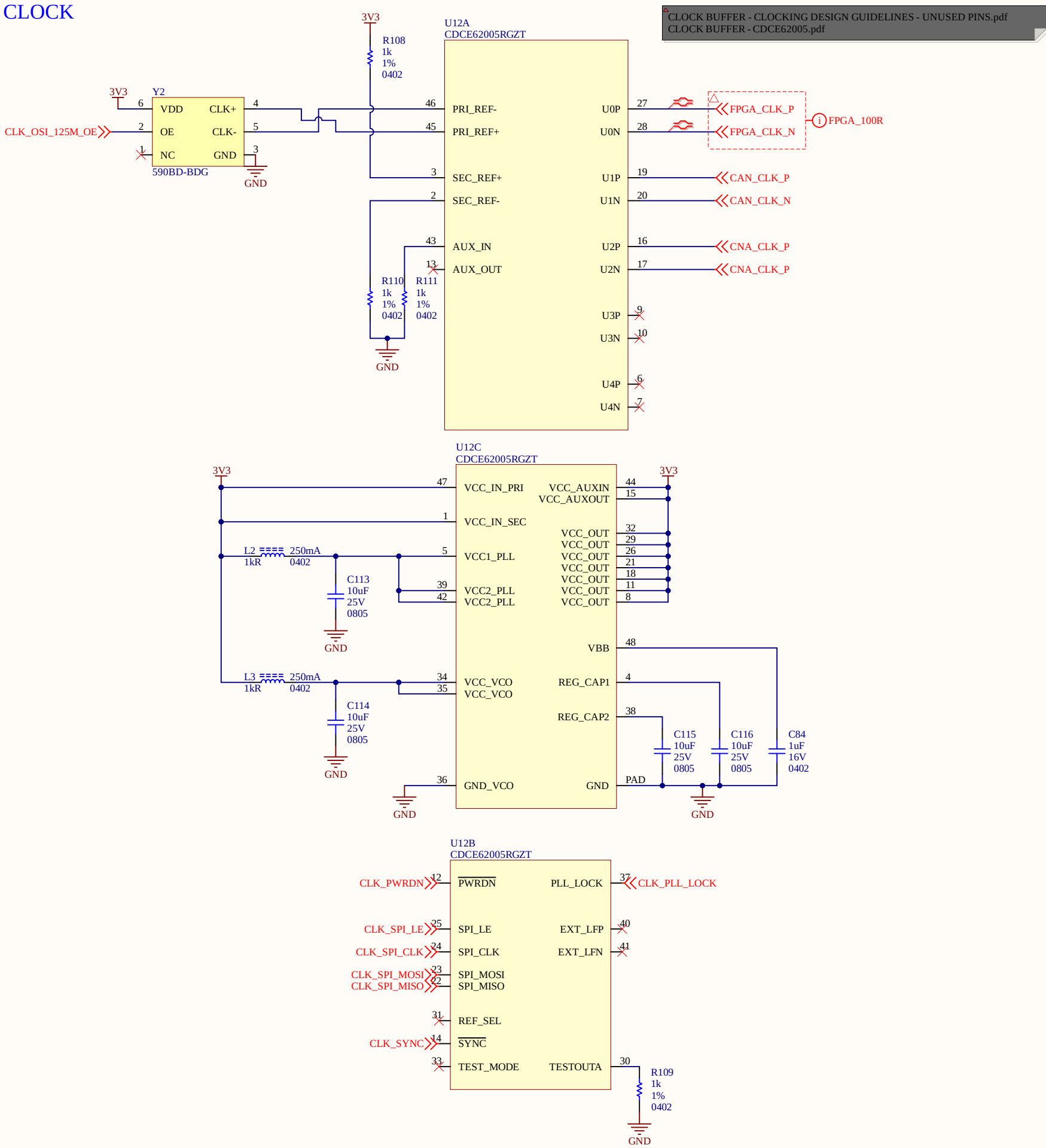


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Global Project			Pleiades		
Size	Group		Revision		
11x17	01		1.00		
Date	13/01/2026		Sheet	1 of	17
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_TITLE.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		



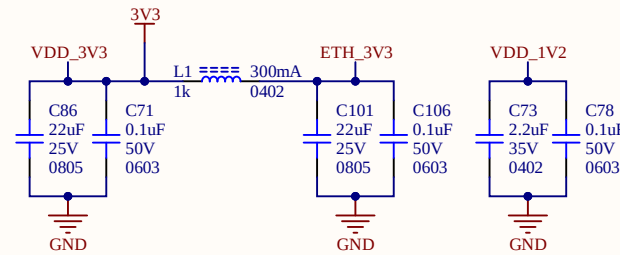
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Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	3 of 17
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_ALIM.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

CLOCK

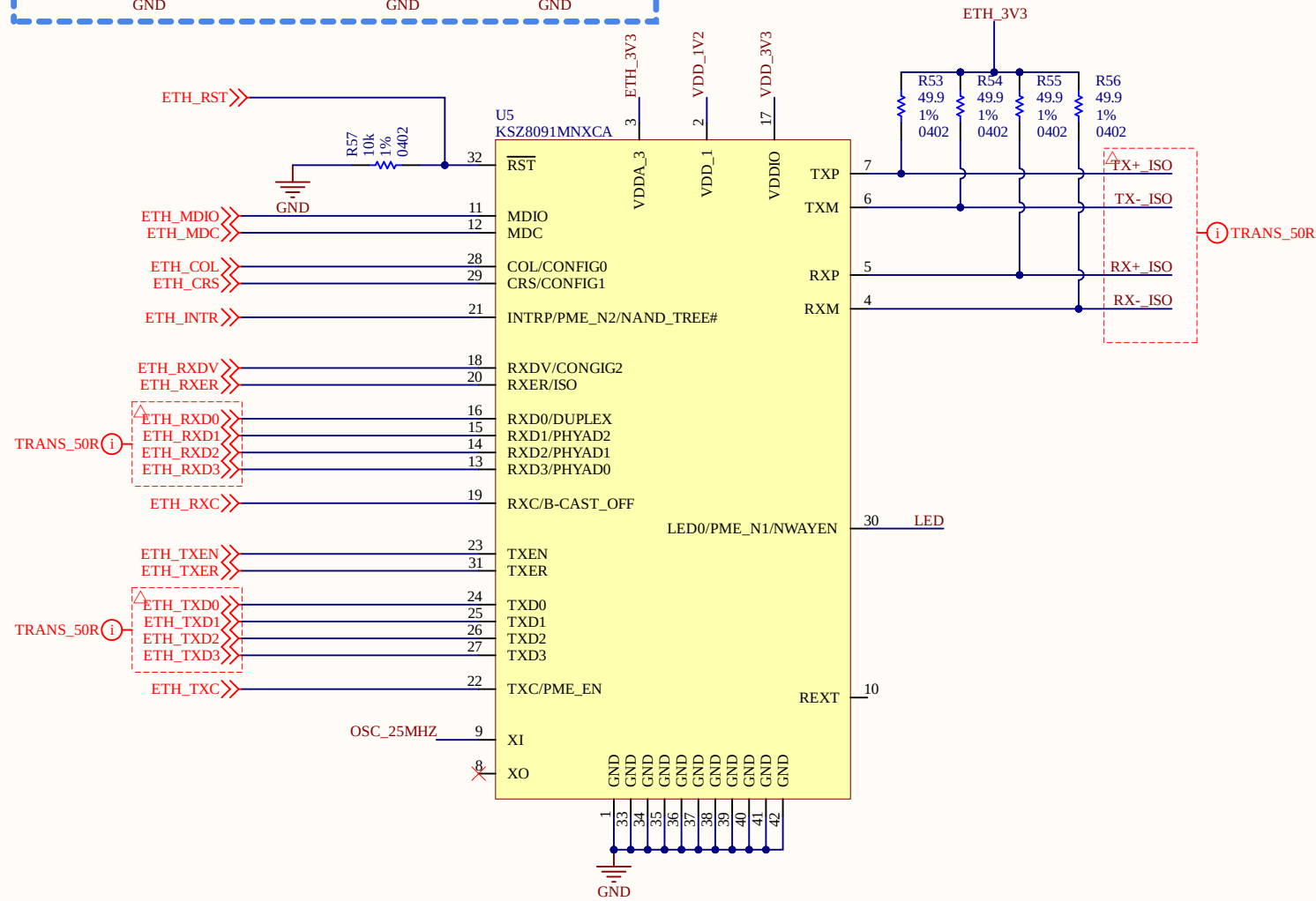


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Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date		Sheet		of	
13/01/2026		4		17	
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_CLOCK.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

ETHERNET ALIMENTATION

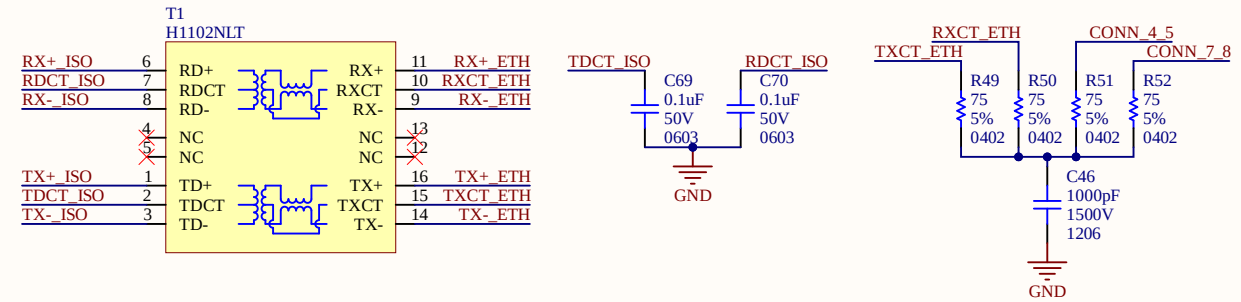


ETHERNET - PHY - KSZ8091MNXCA.pdf

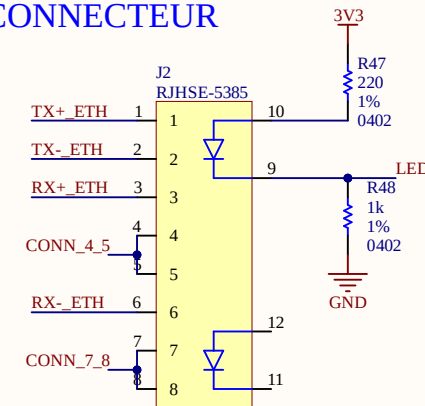


- Isolation du ETHERNET 1.5kV
- Ajout d'un condensateur de 1.5kV
- Ligne de transmission de 50R

ISO

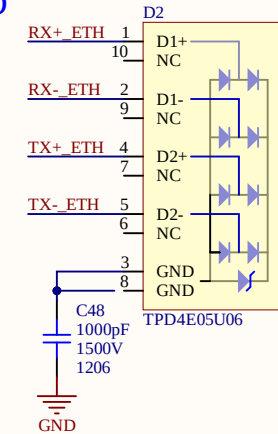


CONNECTEUR

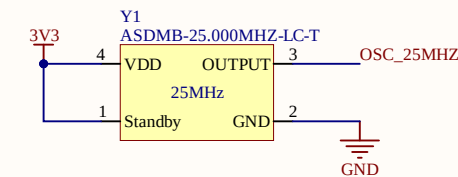


TIA 568A			
Pin #	Wire Color Legend		Signal
1	 White/Green		TX+
2	 Green		TX-
3	 White/Orange		RX+
4	 Blue		TRD2+
5	 White/Blue		TRD2-
6	 Orange		RX-
7	 White/Brown		TR53+
8	 Brown		TRD3-

ESD

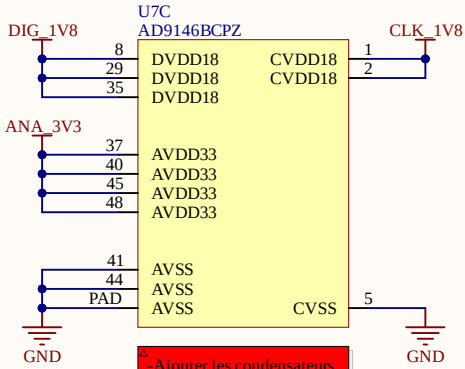
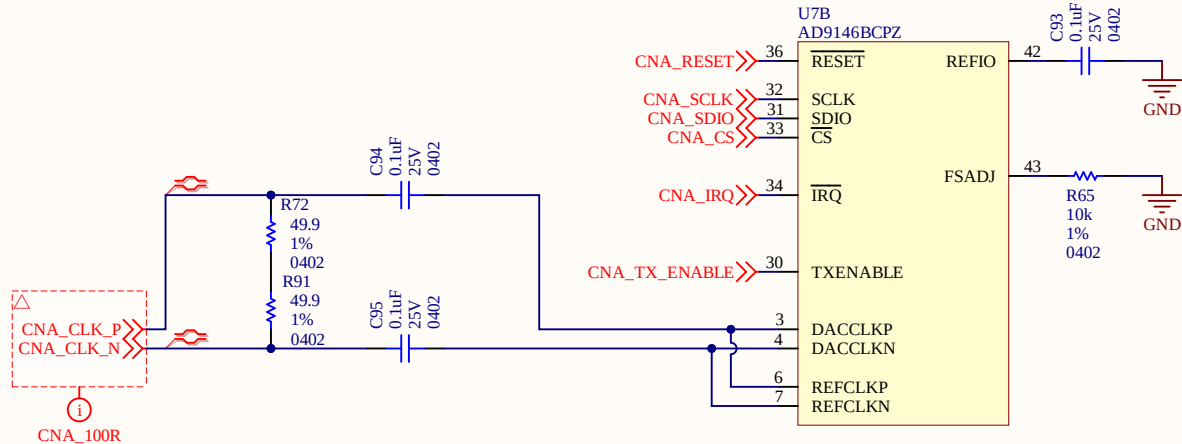
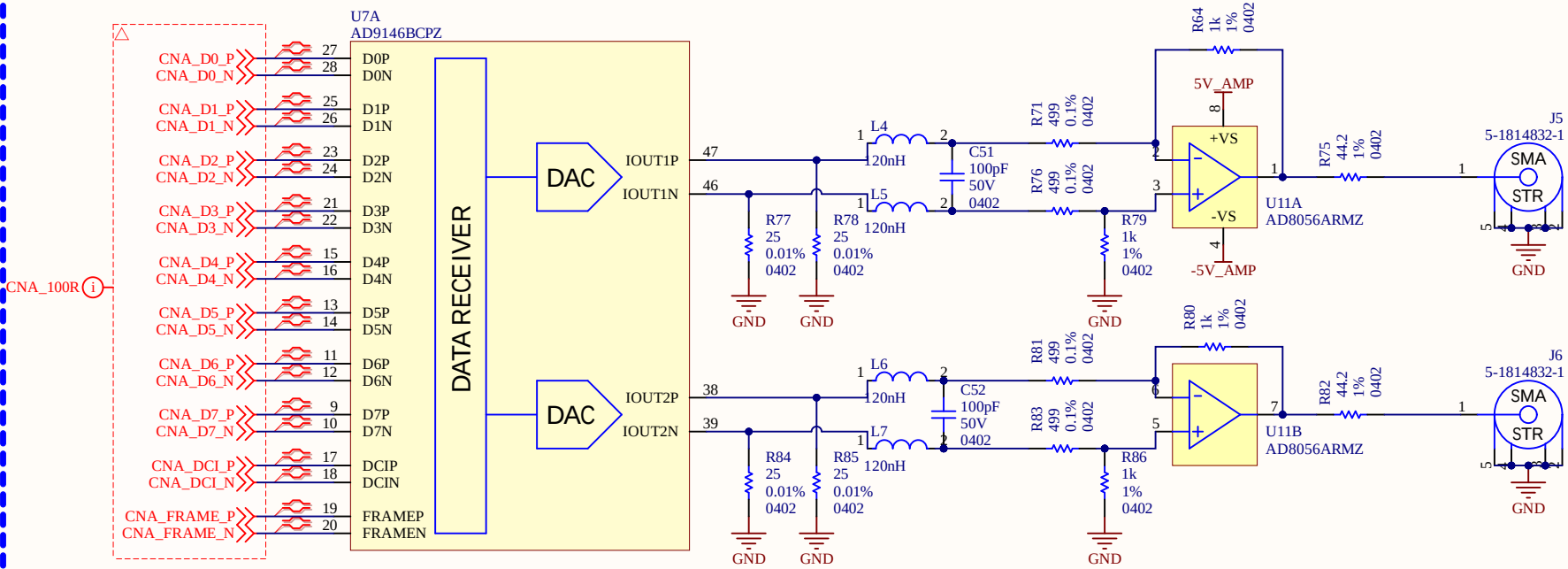


OSCILLATIONS

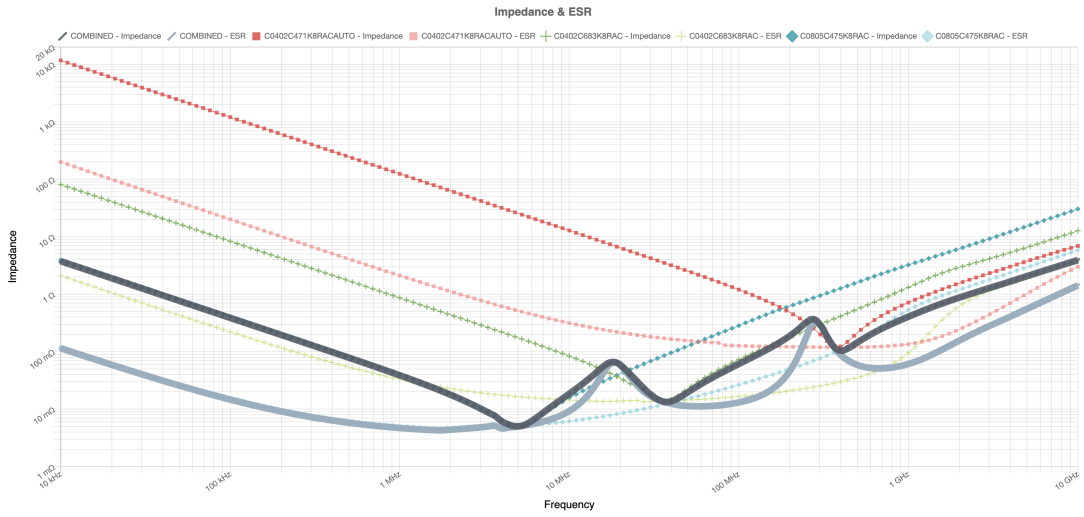


Sheet Name			ETHERNET		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
11x17		01	1.00		
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Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_ETHERNET.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

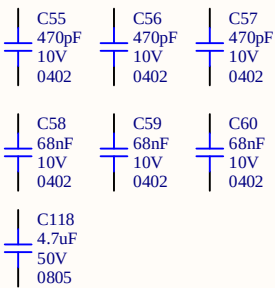
CNA (DAC)



-200mbit/s en DDR LVDS
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4
- Ampli-op aimerait un gain de 2 avec 1V à la sortie, il faut donc 0.5V en entree
-DAC sortie au maximum c'est 20mA, donc avec le 25ohms c'est 0.5V
-On réussi à avoir 1V à la sortie
-5ohms à la sortie à 40MHz donc 45ohms pour adapter
-Filtre passe bas de deuxième ordre (0 à 40MHz), point de coupure 45MHz = L et C



Legend	Tolerance	Part Number	Cap.	V _{DC} *	Dielectric	Qty.	Bias (V)	Amb. (°C)	Add	Remove	
-			Combined	4.3 uF	10	X7R	1	0	25		X
■	10%	C0402C471KBRACAUTO	470 pF	10	X7R	3	0	25	+	×	
+	10%	C0402C683KBRAC	68 nF	10	X7R	3	0	25	+	×	
◆	10%	C0805C475KBRAC	4.7 uF	10	X7R	1	0	25	+	×	



Sheet Name			CNA		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
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11x17	01				1.00
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S7CAS-PLEIADES-CARTE-MERE_P01_CNA.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

CAN (ADC)
DECOUPLAGE

OPAMP - ADA4927-1_ADA4927-2.pdf
P.20

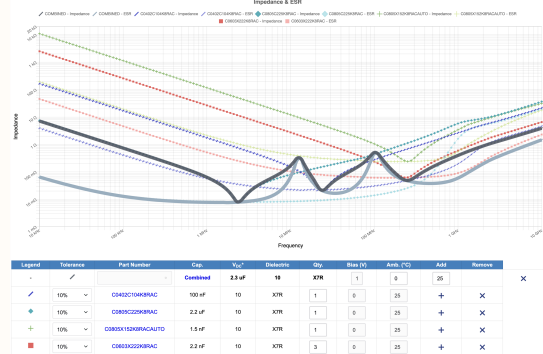
Ajuster les valeurs des résistances

Ajuster les valeurs des résistances

ADC - ADS62P48.pdf
Page 63

Ajouter les condensateurs

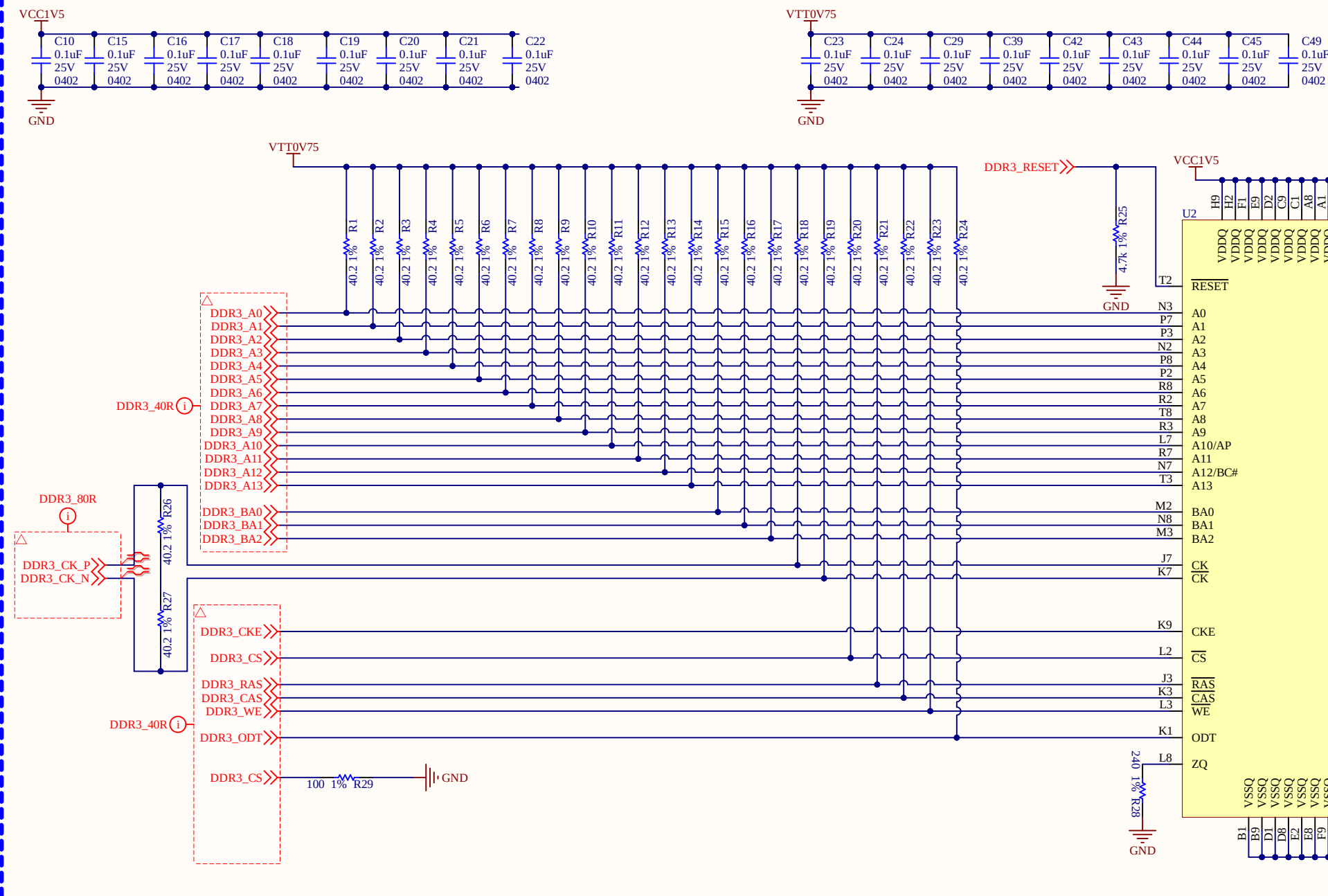
-Choix de OP-AMP pour isolé et pour passer d'analog a digital
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4
- Nous avons pris uniquement AMPLI-OP puisque transfo bloque le DC et on a une composante DC



C62	100nF	C63	2.2uF	C64	1.5nF	C65	2.2nF	C66	2.2nF	C67	2.2nF
10V	0402	10V	0805	10V	0805	10V	0603	10V	0603	10V	0603

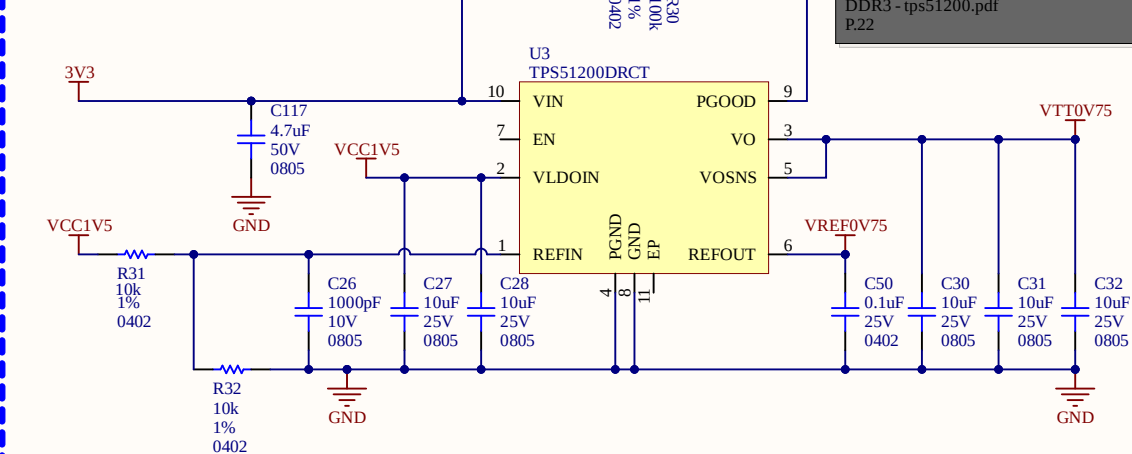
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Global Project			Pleiades		
Size	11x17	Group	01		Revision
Date			13/01/2026		1.00
Date			Sheet	7	of 17
Filename			S7CAS-PLEIADES-CARTE-MERE_P01_CAN.SchDoc		Designers
					Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

FPGA



ANALYSER LES COURANTS POUR LE VTT
AJUSTER LES CONDENSATEURS

ALIM DDR3



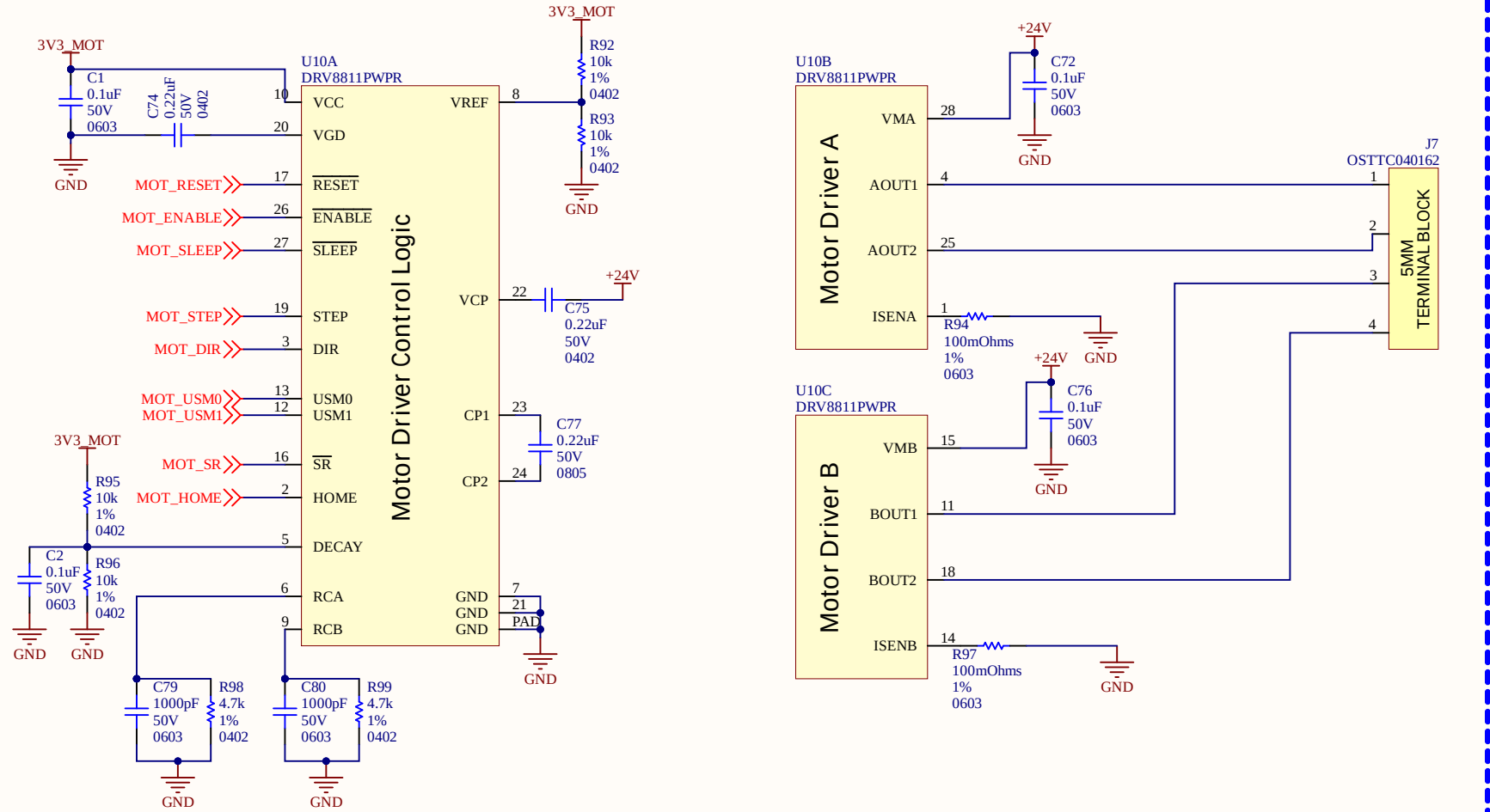
-Ligne de transmission de 40R
-Resistance parallèle sur la ligne de transmission de 40.2Ohms
-Clock en pair différentiel 80R

Sortie PGOOD?
EN?

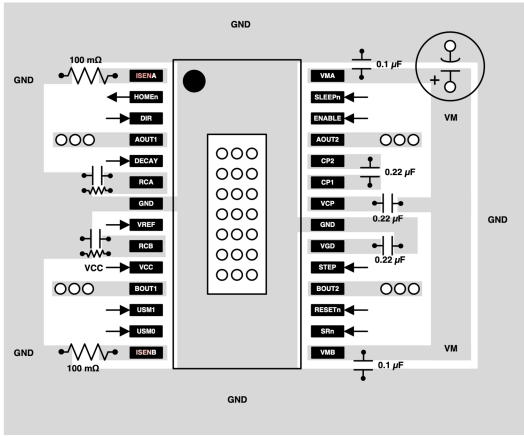
ACT	33.4 mW
RD	21.0 mW
WR	23.4 mW
READ I/O	32.7 mW
Write ODT	136.1 mW
Total RD/WR/Term Power	213.2 mW
ACT_STBY	14.9 mW
PRE_STBY	3.1 mW
ACT_PDN	2.9 mW
PRE_PDN	0.7 mW
REF	4.2 mW
Total Background Power	25.8 mW
Total DDR3 SDRAM Power	272.3 mW
TERM 2nd rank	0.0 mW

Sheet Name			
DDR3			
Project Title			
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13/01/2026		8	17
Filename		Designers	
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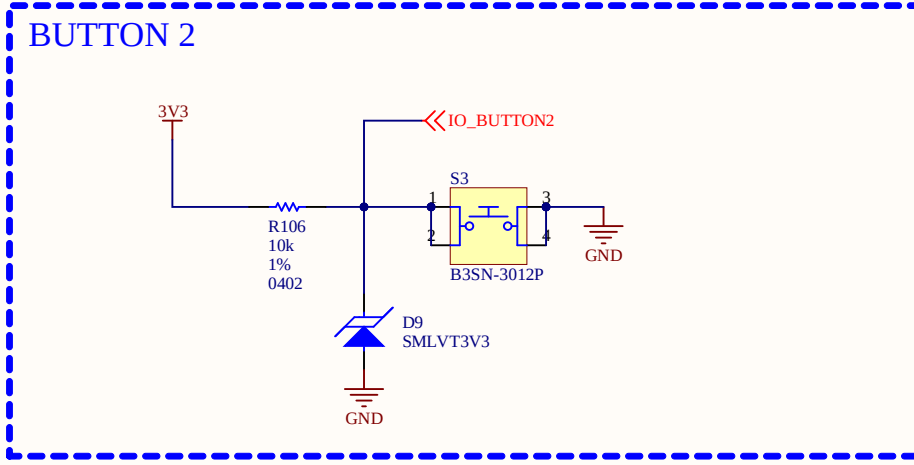
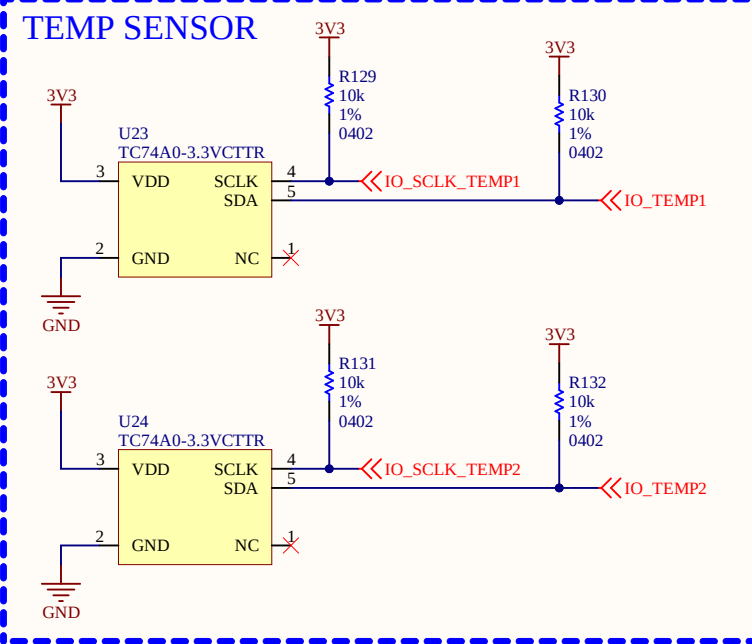
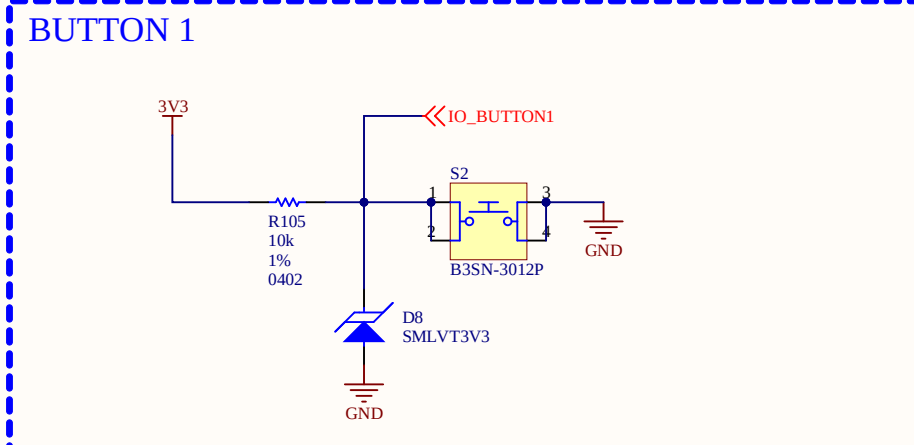
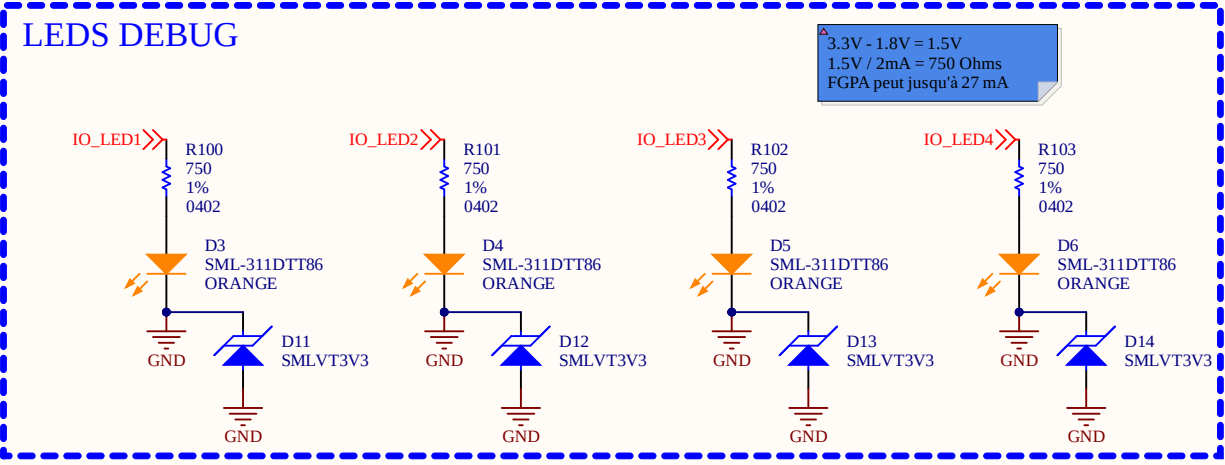
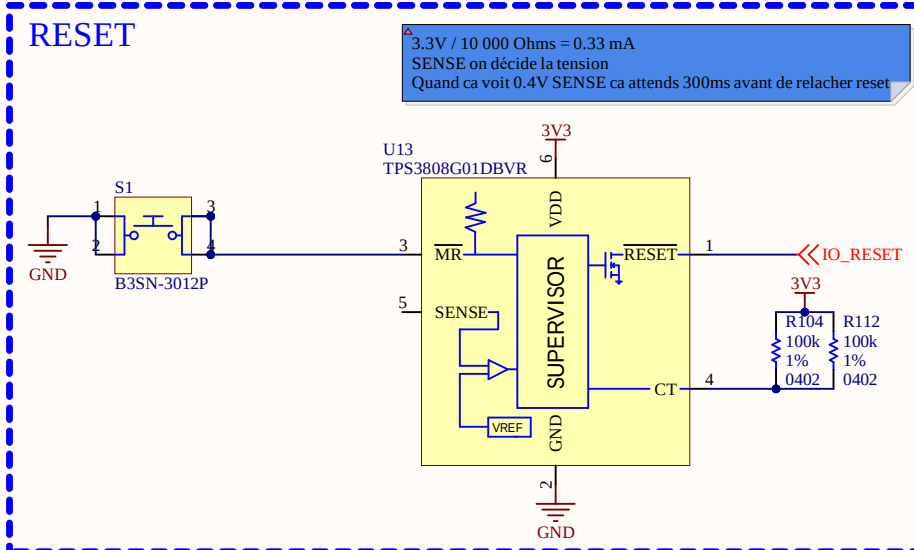
MOTOR



-Tous sur la BANK1
-ISENS = $1.65/(8 \times 1.9A) = 0,108$ Ohms

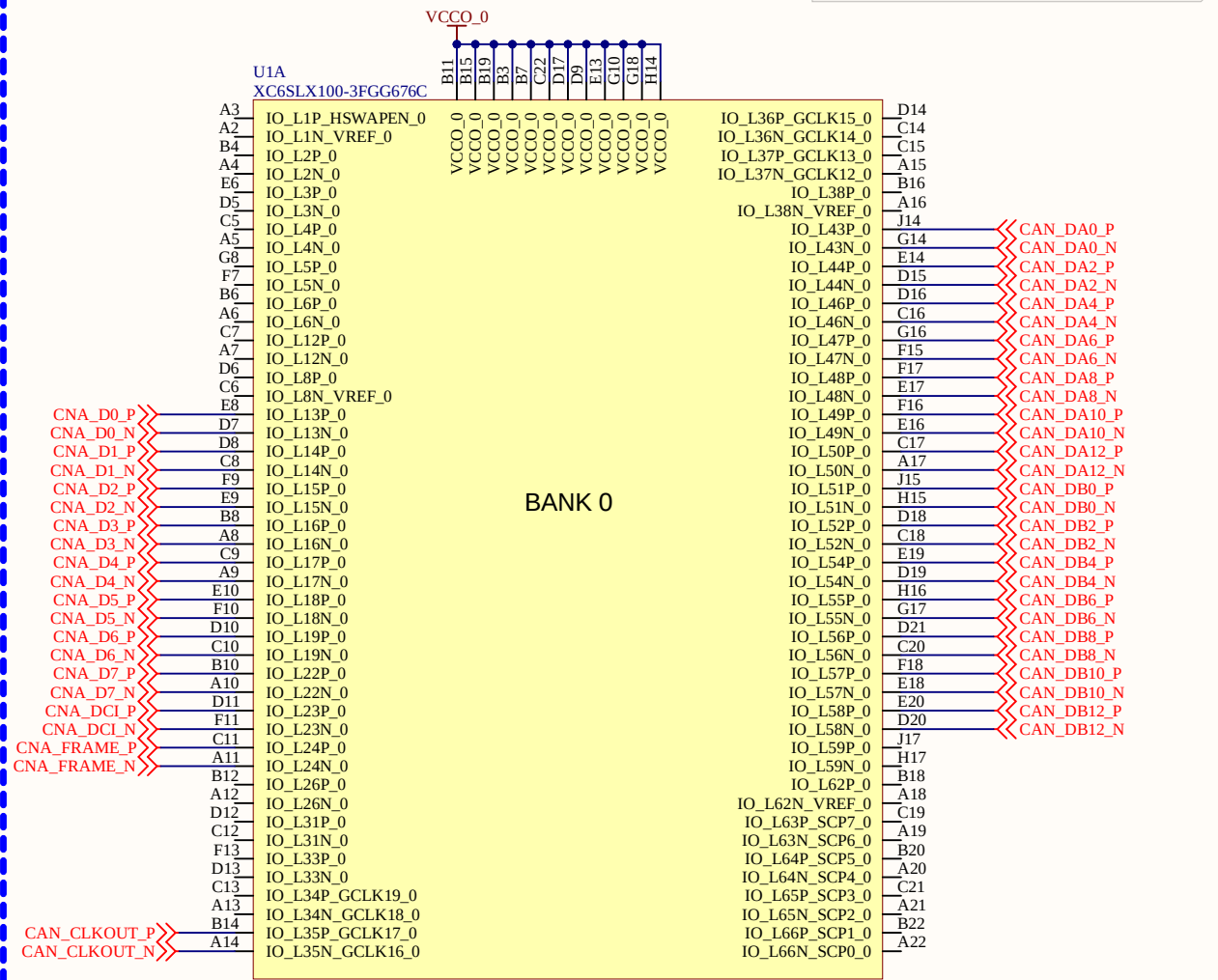


Sheet Name		MOTORS	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
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S7CAS-PLEIADES-CARTE-MERE_P01_MOTORS.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	



Sheet Name			IO		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
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11x17	01		1.00		
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Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_IO.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA



IO CAN est LVDS donc il faut DIFF_TERM = "TRUE"

- LVDS seulement sur BANK0 et BANK2
- Toute les CAN et CNA sont mit sur cette BANK

Sheet Name			FPGA - BANK 0		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date			Sheet		
13/01/2026			11 of 17		
Filename				Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK0.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

FPGA

VCCO_1

U1B
XC6SLX100-3FGG676C

AB23
AD25
M20
P23
T25
U18
V21
W23
Y25

L19
K19
L23
L24
P20
N21
M23
N24
L17
K18
P24
P26
M19
L18
R25
R26
M18
N19
N22
N23
N17
N18
R23
R24
N20
M21
P21
P22
V23
W24
U25
U26
W25
W26
V24
V26
T24
T26
Y24
Y26
AD24
AD26
AB24
AB26
AC25
AC26

IO_L28P_1
IO_L28N_VREF_1
IO_L29P_A23_M1A13_1
IO_L29N_A22_M1A14_1
IO_L30P_A21_M1RESET_1
IO_L30N_A20_M1A11_1
IO_L31P_A19_M1CKE_1
IO_L31N_A18_M1A12_1
IO_L32P_A17_M1A8_1
IO_L32N_A16_M1A9_1
IO_L33P_A15_M1A10_1
IO_L33N_A14_M1A4_1
IO_L34P_A13_M1WE_1
IO_L34N_A12_M1BA2_1
IO_L35P_A11_M1A7_1
IO_L35N_A10_M1A2_1
IO_L36P_A9_M1BA0_1
IO_L36N_A8_M1BA1_1
IO_L37P_A7_M1A0_1
IO_L37N_A6_M1A1_1
IO_L38P_A5_M1CLK_1
IO_L38N_A4_M1CLKN_1
IO_L39P_M1A3_1
IO_L39N_M1ODT_1
IO_L40P_GCLK11_M1A5_1
IO_L40N_GCLK10_M1A6_1
IO_L41P_GCLK9_IRDY1_M1RASN_1
IO_L41N_GCLK8_M1CASN_1
IO_L42P_GCLK7_M1UDM_1
IO_L42N_GCLK6_TRDY1_M1LDM_1
IO_L43P_GCLK5_M1DQ4_1
IO_L43N_GCLK4_M1DQ5_1
IO_L44P_A3_M1DQ6_1
IO_L44N_A2_M1DQ7_1
IO_L45P_A1_M1LDQS_1
IO_L45N_A0_M1LDQSN_1
IO_L46P_FCS_B_M1DQ2_1
IO_L46N_FOE_B_M1DQ3_1
IO_L47P_FWE_B_M1DQ0_1
IO_L47N_LDC_M1DQ1_1
IO_L48P_HDC_M1DQ8_1
IO_L48N_M1DQ9_1
IO_L49P_M1DQ10_1
IO_L49N_M1DQ11_1
IO_L50P_M1UDQS_1
IO_L50N_M1UDQSN_1

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VCCO_1
VCCO_1
VCCO_1
VCCO_1
VCCO_1
VCCO_1
VCCO_1

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IO_L51N_M1DQ13_1
IO_L52P_M1DQ14_1
IO_L52N_M1DQ15_1
IO_L53P_1
IO_L53N_VREF_1
IO_L55P_1
IO_L55N_1
IO_L56P_1
IO_L56N_1
IO_L57P_1
IO_L57N_1
IO_L58P_1
IO_L58N_1
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IO_L72P_1
IO_L72N_1
IO_L73P_1
IO_L73N_1
IO_L74P_AWAKE_1
IO_L74N_DOUT_BUSY_1

AA25
AA26
AE25
AE26
T23
U24
R22
R21
P17
P18
R20
R19
R17
R18
T22
U23
T18
T19
U21
U22
U17
V17
AA23
AA24
T20
U20
AC23
AC24
V18
V19
AE24
AF25
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Y22
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AE23
AF24

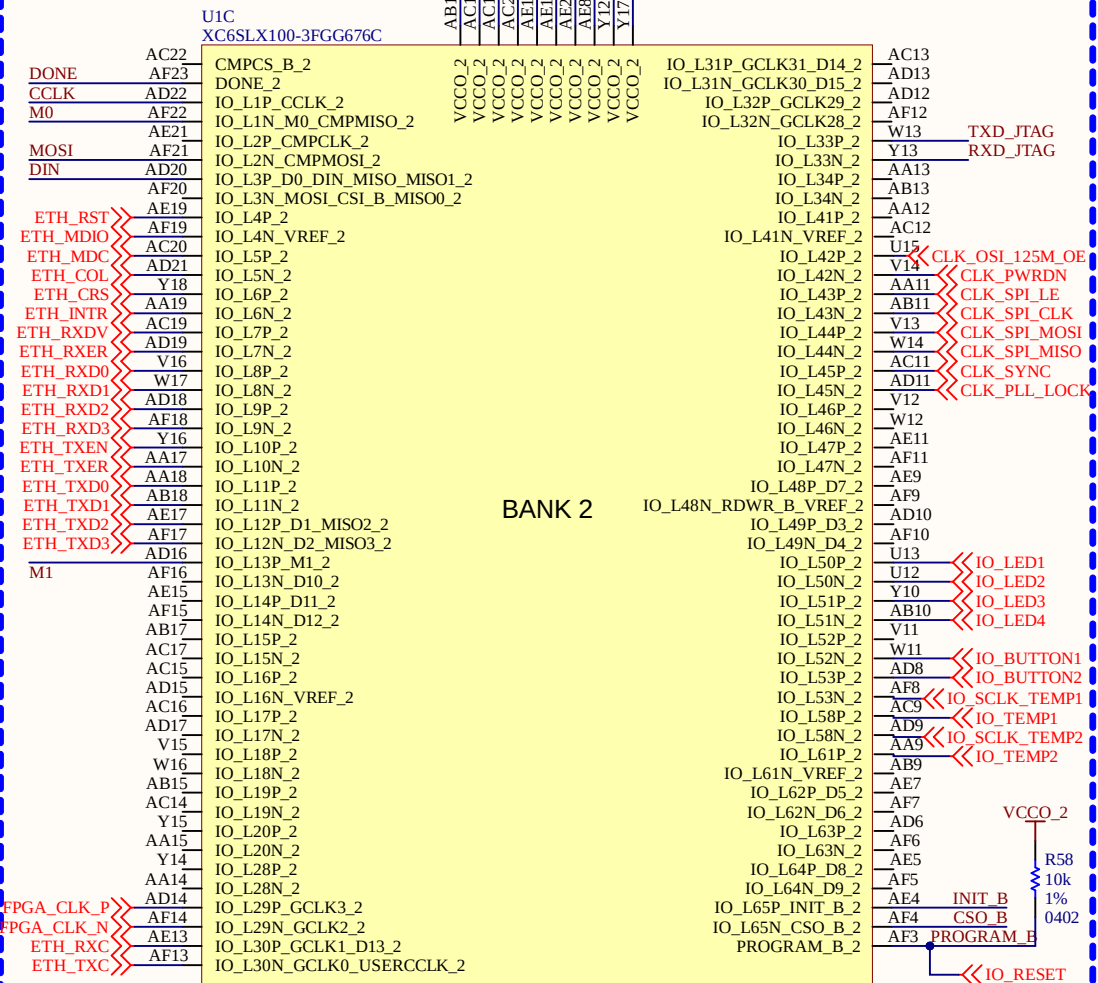
MOT_HOME
MOT_SR
MOT_USM1
MOT_USM0
MOT_DIR
MOT_STEP
MOT_SLEEP
MOT_ENABLE
MOT_RESET

BANK 1

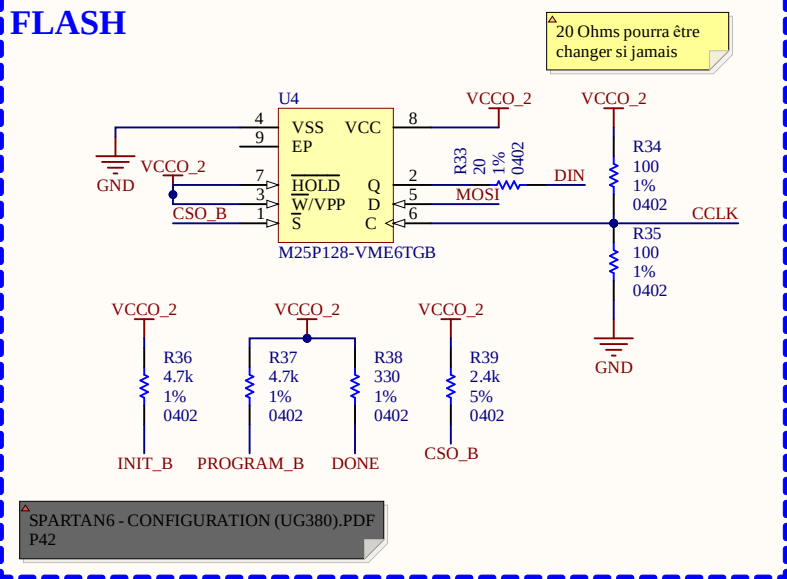
-Toute les controles de moteur sont sur cette BANK

Sheet Name			FPGA - BANK 1		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
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11x17	01	1.00			
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S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK1.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

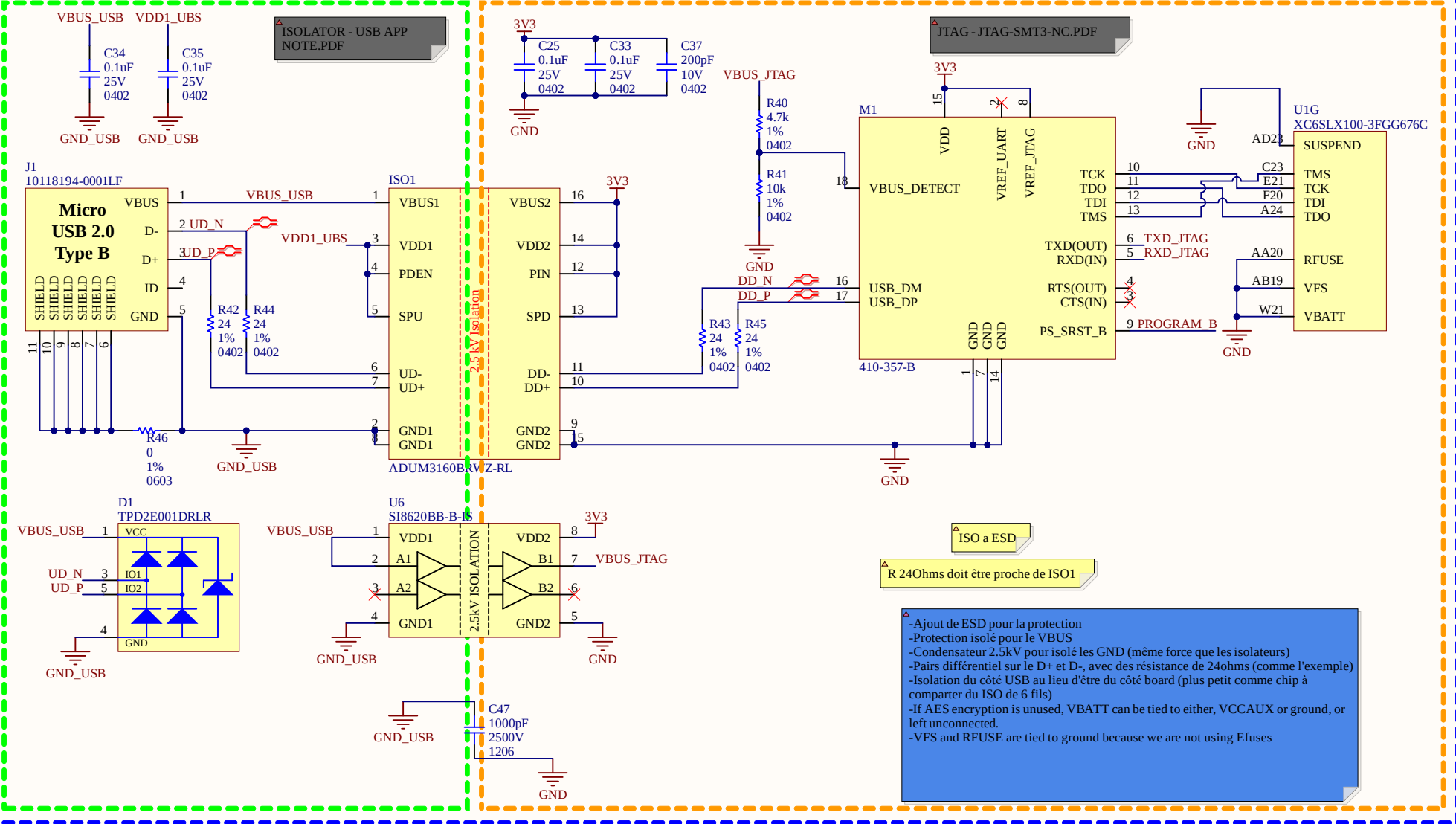
FPGA



FLASH



USB



FPGA - BANK 2

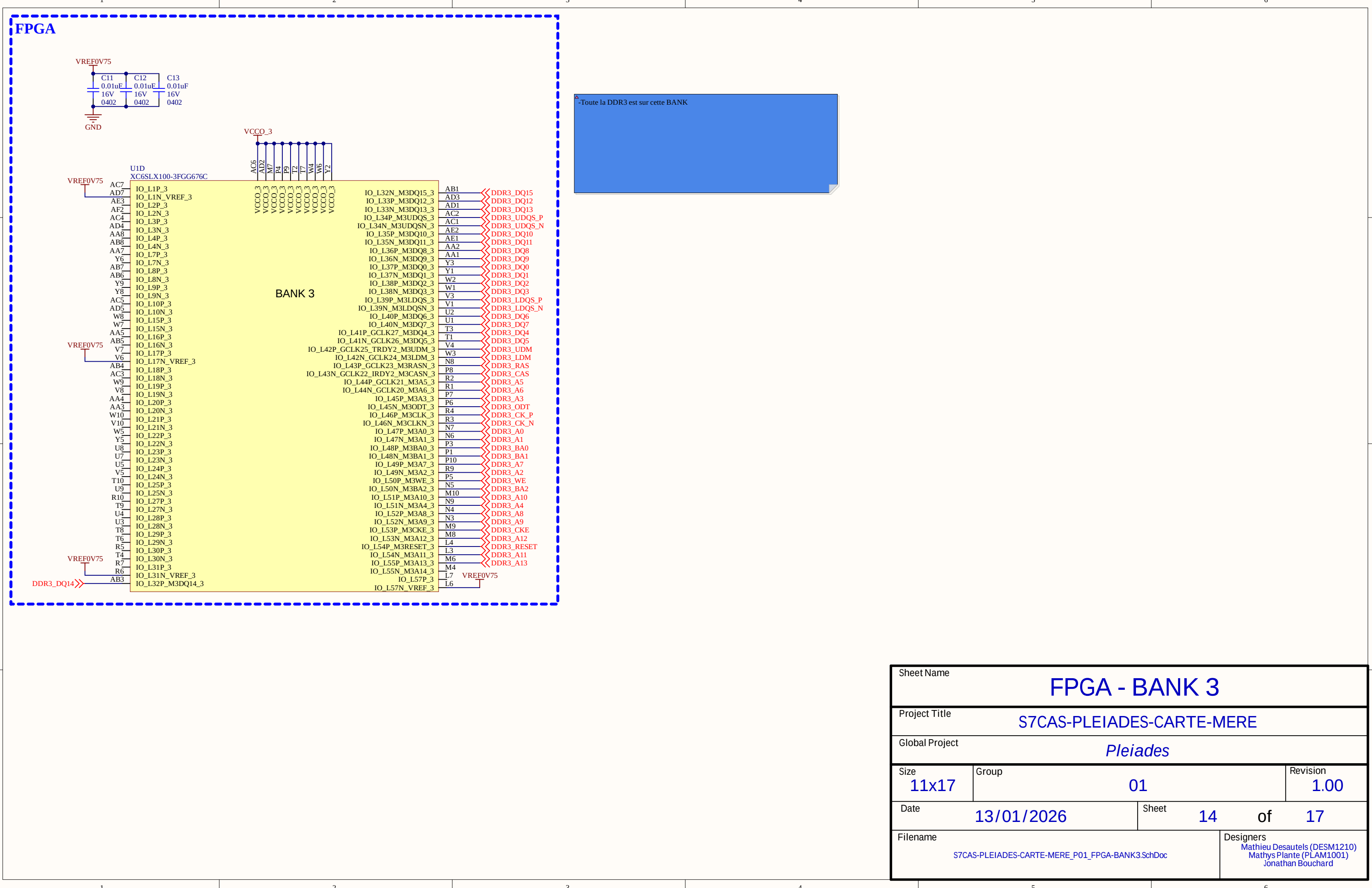
S7CAS-PLEIADES-CARTE-MERE

Pleiades

Size 11x17 Group 01 Revision 1.00

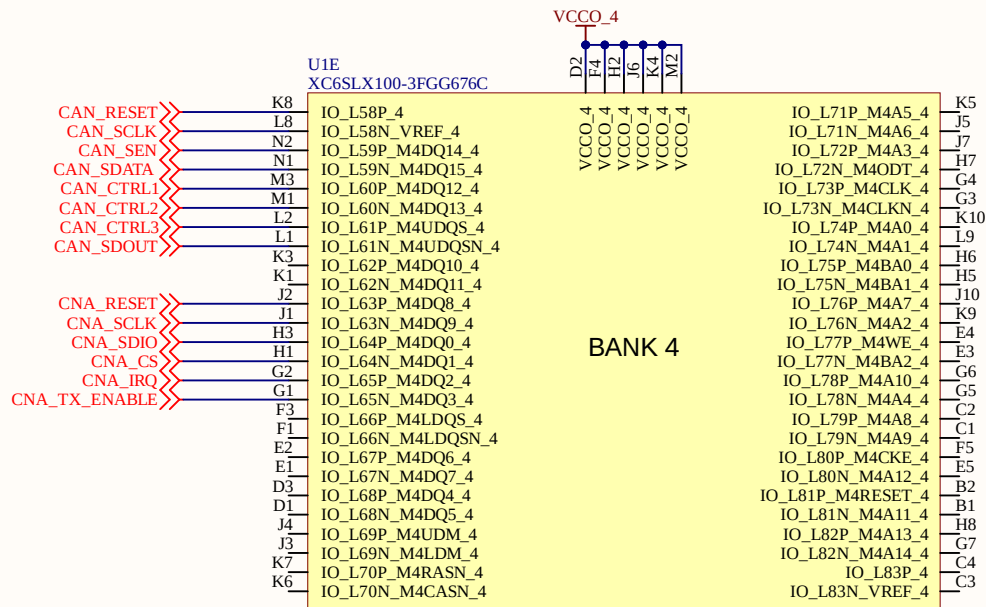
Date 13/01/2026 Sheet 13 of 17

Filename S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK2.SchDoc Designers Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard



Sheet Name			FPGA - BANK 3		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
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Filename				Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK3.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

FPGA



- Toute le controle du CNA et CAN sont sur cette BANK
- Leur tension n'est pas la même que celle de leur données

Sheet Name			FPGA - BANK 4		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group	Revision			
11x17	01	1.00			
Date	13/01/2026	Sheet	15	of	17
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK4.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

